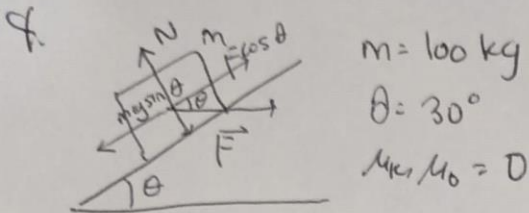


TUGAS 02 F1101 MODUL 02

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Jawaban ($g = 9,8 \text{ m/s}^2$)



(a) $\sum F_x = 0$

$$\Rightarrow F \cos \theta - mg \sin \theta = 0 \Rightarrow F = \frac{mg \sin \theta}{\cos \theta}$$

$$\Rightarrow F = mg \cdot \tan \theta$$

$$= 100 \cdot 9,8 \cdot \tan 30^\circ$$

$$= 980 \cdot \frac{1}{3}\sqrt{3}$$

$$\approx \boxed{565,80326 \text{ N}}$$

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(b) $\sum F_y = 0$

$$\Rightarrow N - mg \cos \theta - F \sin \theta = 0$$

$$\Rightarrow N = mg \cos \theta + F \sin \theta$$

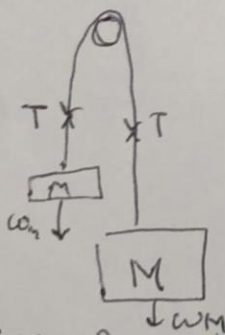
$$= 980 \cdot \cos(30^\circ) + 980 \cdot \frac{1}{3}\sqrt{3} \cdot \sin(30^\circ)$$

$$= 980 \cdot \frac{1}{2}\sqrt{3} + \frac{980}{2} \cdot \frac{1}{3}\sqrt{3}$$

$$\approx \boxed{1131,60652 \text{ N}}$$

~

5.



$m_m = 10 \text{ kg}$, $m = \text{monyet}$

$m_M = 15 \text{ kg}$ $M = \text{peti}$

"katrol" tak bermassa

(a) Agar bisa mengangkat, gaya yang diberikan monyet = peti

$$\Rightarrow F_{\text{monyet}} = F_{\text{peti}}$$

$$\Rightarrow m_m \cdot a_{\text{min}} + m_m \cdot g = m_M \cdot g$$

$$\Rightarrow a_{\text{min}} = \frac{m_M \cdot g - m_m \cdot g}{m_m} = \frac{15 \cdot 9,8 - 10 \cdot 9,8}{10} = \boxed{4,9 \text{ m/s}^2}$$

~

$$(b) T - w_m = m_m \cdot a_m \quad (1)$$

$$\Rightarrow w_M - T = m_M \cdot a_M \quad (2)$$

Karena satu sistem, $a_m = a_M$

$$\Rightarrow \frac{T - w_m}{m_m} = \frac{w_M - T}{m_M}$$

$$\Rightarrow \frac{T - 98}{10} = \frac{159.8 - T}{15} \Rightarrow T = 117.6 \text{ N}$$

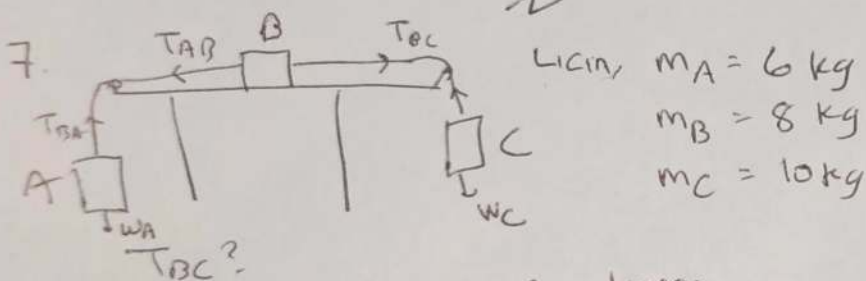
Dari persamaan (1), $a_m = \frac{T - w_m}{m_m}$

$$= \frac{117.6 - 98}{10}$$

$$= \boxed{1.96 \text{ m/s}^2}$$

(c) Karena $m_m < m_M$, manjet bergerak ke atas,
maka percepatan ke atas $\uparrow$

(d) Menurut (b), $T = 117.6 \text{ N}$



Karena $w_C > w_A$, sistem ke kanan

$$w_C - T_{BC} = m_C \cdot a \quad (i)$$

$$\frac{w_C - w_A}{m_A + m_B + m_C} = a \quad (ii)$$

$$m_A + m_B + m_C$$

$$\Rightarrow a = \frac{98 - 58.8}{24} = 1.633 \text{ m/s}^2$$

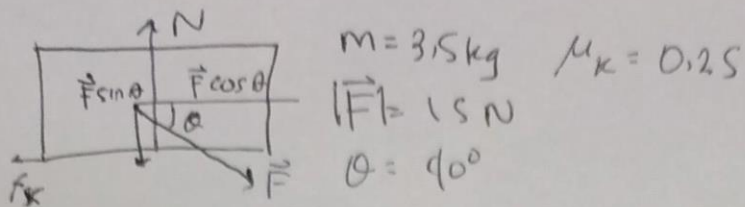
Substitusi ke persamaan (i), didapat

$$98 - T_{BC} = 10 \cdot 1.633$$

$$\Rightarrow T_{BC} = 98 - 16.33$$

$$= \boxed{81.66 \text{ N}}$$

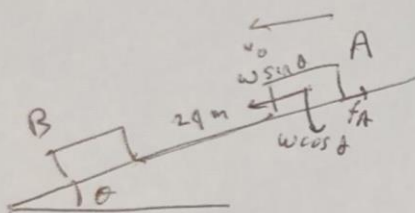
9.



$$\begin{aligned}
 \text{(a)} \quad f_k &= \mu_k \cdot N \\
 &= 0.25 \cdot (mg + F \sin \theta) \\
 &= 0.25 \cdot (3.5 \cdot 9.8 + 15 \cdot \sin 40^\circ) \\
 &\approx 0.25 (43.9918) \\
 &\approx \boxed{10.9854 \text{ N}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad ma &= F \cos \theta - f_k \\
 \Rightarrow a &= \frac{F \cos \theta - f_k}{m} \\
 &\approx \frac{15 \cdot \cos 40^\circ - 10.9854}{3.5} \\
 &\approx \boxed{0.1442 \text{ m/s}^2}
 \end{aligned}$$

10.



$$\begin{aligned}
 \theta &= 12^\circ \\
 d &= 29 \text{ m} \\
 v_0 &= 18.0 \text{ m/s}
 \end{aligned}$$

$$\begin{aligned}
 \text{(a)} \quad \mu_k &= 0.6 \\
 f_A &= \mu_k \cdot N \\
 &= 0.6 (m_A g \cdot \cos \theta) \\
 &= 0.6 \cdot m_A \cdot 9.8 \cdot \cos 12^\circ \\
 &\approx 5.7515 m_A
 \end{aligned}$$

$$\begin{aligned}
 w \sin \theta - f_A &= m_A \cdot a \\
 \Rightarrow a &= \frac{w \sin \theta - f_A}{m_A} \\
 &= \frac{9.8 m_A \cdot \sin 12^\circ - 5.7515 m_A}{m_A} \\
 &\approx -3.71 \text{ m/s}^2
 \end{aligned}$$

$$\begin{aligned}
 v_f^2 &= v_0^2 + 2ad \\
 \Rightarrow v_f &= \sqrt{324 - 178.08} \\
 &= \sqrt{145.92} \\
 &\approx \boxed{12.08 \text{ m/s}}
 \end{aligned}$$

$$\text{(b)} \quad \mu_k = 0.1$$

Pakai langsung mencari percepatan

$$\begin{aligned}
 a &= g (\sin \theta - \mu_k \cdot \cos \theta) \\
 &= 9.8 (\sin 12^\circ - 0.1 \cdot \cos 12^\circ) \\
 &\approx 9.8 \cdot 0.1101 \\
 &\approx 1.08 \text{ m/s}^2
 \end{aligned}$$

$$\begin{aligned}
 v_f^2 &= v_0^2 + 2ad \\
 \Rightarrow v_f &= \sqrt{324 + 51.84} \\
 &= \sqrt{375.84} \\
 &\approx \boxed{19.38 \text{ m/s}}
 \end{aligned}$$